

Global AI & Tech Layoffs (2020-2026): An Explanatory Data Analysis

Course :

M512 Data Visualisation and Communication

Name :

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Dataset Link :

<https://www.kaggle.com/datasets/prajitdatta/global-ai-and-tech-layoffs-dataset-20202026>

Audience:

HR strategy leads, workforce planning teams, and investors at a mid-size tech company evaluating hiring/layoff risk.

Goal:

Will aim to make the audience know why, where and when tech layoffs occur — and how much is being caused by AI automation and how much is caused by traditional economic cycles. This provides data for workforce planning, investor risk assessment, as well as strategic shifts in AI adoption.

Data:

1850 layoffs from AI/tech companies across the globe from 2020 to 2026 with company metadata (industry, size, status), layoff data (numbers, % of employees, reason, departments impacted), and market reaction (changes to stock prices).

Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Code

```
df = pd.read_csv("global_ai_tech_layoffs_2020_2026.csv")

print(f"Shape: {df.shape}")
df.head()
```

Output

Shape: (1850, 22)

	event_id	company	industry	hq_country	hq_city	continent	founded_year	company_size_est	company_status	date_announced	...	quarter	number_laid_off	percentage_of_workforce	departments_affected	primary_reason	impact_scope	severity_info	source	stock_price_change_1wk_pct	is_related_layoff
0	LO-1034	Swiggy	Food Delivery	India	Bangalore	Asia	2014	6000	Private	2020-01-04	...	Q1	722	12.0	HR / People; Legal; Product	Post-Pandemic Overhiring Correction	Global	Varies by region	News Report - The Verge	NaN	False
1	LO-1020	Oracle	Enterprise Software	United States	Austin	North America	1977	164000	Public	2020-01-06	...	Q1	6189	3.8	Legal; Business Development; Data Science / AI ...	Investor Pressure	Global	6 months base salary	News Report - Bloomberg	-1.30	False
2	LO-1051	Tesla	Electric Vehicles / AI	United States	Austin	North America	2003	140000	Public	2020-01-08	...	Q1	9512	6.8	Operations	Product Shutdown	Multiple Regions	6 months base salary	News Report - CNBC	-3.89	False
3	LO-1063	Anthropic	Artificial Intelligence	United States	San Francisco	North America	2021	1500	Private	2020-01-09	...	Q1	132	8.8	Marketing; Sales; Research; Customer Support	AI Automation Replacing Roles	United States	Varies by region	Company Press Release	NaN	True

	event_id	company	industry	hq_country	hq_city	continent	founded_year	company_size_est	company_status	date_announced	...	quarter	number_laid_off	percentage_of_workforce	departments_affected	primary_reason	impact_scope	severance_info	source	stock_price_change_1wk_pct	is_ai_related_layoff
4	LO-1139	Alibaba	E-Commerce / Cloud	China	Hangzhou	Asia	1999	220000	Public	2020-01-11	...	Q1	16245	7.4	Recruiting; Supply Chain; Legal; Engineering	Merger / Acquisition	China	4 weeks base salary	News Report - TechCrunch	-5.16	False

Data Exploration

Before visualizing insights, we need to understand the shape, quality, and quirks of this dataset — including missing values, inconsistent categories, and outliers.

Code

```
print("Data types:\n", df.dtypes)
print("\nMissing values:\n", df.isna().sum()[df.isna().sum() > 0])
print(f"\nDuplicate rows: {df.duplicated().sum()}")
print(f"\nDate range: {df['date_announced'].min()} to {df['date_announced'].max()}")
```

Output

```
Data types:
event_id          object
company           object
industry          object
hq_country        object
hq_city           object
continent         object
founded_year      int64
company_size_est  int64
company_status    object
date_announced   object
year              int64
month             int64
quarter           object
number_laid_off   int64
percentage_of_workforce float64
departments_affected object
primary_reason    object
impact_scope      object
severance_info    object
source            object
stock_price_change_1wk_pct float64
is_ai_related_layoff bool
dtype: object
```

```
Missing values:
stock_price_change_1wk_pct    461
dtype: int64
```

```
Duplicate rows: 0
```

```
Date range: 2020-01-04 to 2026-02-28
```

Code

```
print("Unique industries:", df['industry'].nunique())
print(df['industry'].value_counts().head(20))

print("\nUnique primary reasons:", df['primary_reason'].nunique())
print(df['primary_reason'].value_counts())
```

```
print("\nContinents:", df['continent'].unique())
print("\nCompany status:", df['company_status'].unique())
```

Output

Unique industries: 49

```
industry
Fintech          143
Social Media     106
E-Commerce       97
Artificial Intelligence  91
Enterprise Software  74
Food Delivery    71
Semiconductors   68
Design Software  61
IT Services      60
Travel Tech      57
E-Commerce / Cloud  52
Consumer Electronics  49
Ride-Hailing     43
Cloud Communications  33
Streaming        33
Social Media / AI  31
Cryptocurrency   31
Real Estate Tech  29
Generative AI    29
Electric Vehicles / AI  29
Name: count, dtype: int64
```

Unique primary reasons: 17

```
primary_reason
AI Automation Replacing Roles    203
Restructuring                   181
Shift to AI-First Strategy       177
Failed Expansion                 108
Regulatory Pressure             108
Duplicate Roles After Merger     106
Market Downturn                 101
Post-Pandemic Overhiring Correction  95
Investor Pressure                93
Product Shutdown                91
Profitability Push              90
Cost Reduction                  88
Economic Uncertainty            88
Merger / Acquisition            86
Return-to-Office Attrition      81
Strategic Pivot                 78
Declining Revenue               76
Name: count, dtype: int64
```

Continents: ['Asia' 'North America' 'Europe' 'Oceania' 'South America']

Company status: ['Private' 'Public']

Code

```
print(df[['number_laid_off', 'percentage_of_workforce', 'company_size_est',
          'stock_price_change_1wk_pct']].describe())
```

```
fig, axes = plt.subplots(1, 3, figsize=(16, 4))
sns.histplot(df['number_laid_off'], bins=40, ax=axes[0])
axes[0].set_title('Number Laid Off (highly skewed)')
sns.histplot(df['percentage_of_workforce'], bins=30, ax=axes[1])
axes[1].set_title('% of Workforce Laid Off')
sns.histplot(df['stock_price_change_1wk_pct'].dropna(), bins=30, ax=axes[2])
axes[2].set_title('Stock Price Change 1wk (%)')
plt.tight_layout()
plt.show()
```

Output

	number_laid_off	percentage_of_workforce	company_size_est \
count	1850.000000	1850.000000	1.850000e+03
mean	4452.665946	6.052000	7.651009e+04
std	12196.345154	4.307832	1.852442e+05
min	10.000000	0.300000	8.000000e+01
25%	197.250000	2.800000	5.000000e+03
50%	657.500000	4.900000	1.100000e+04
75%	3862.000000	8.200000	8.490000e+04
max	205254.000000	25.500000	1.540000e+06

	stock_price_change_1wk_pct
count	1389.000000

```

mean          -1.994867
std           3.986245
min          -15.890000
25%          -4.700000
50%          -2.000000
75%           0.730000
max           10.250000

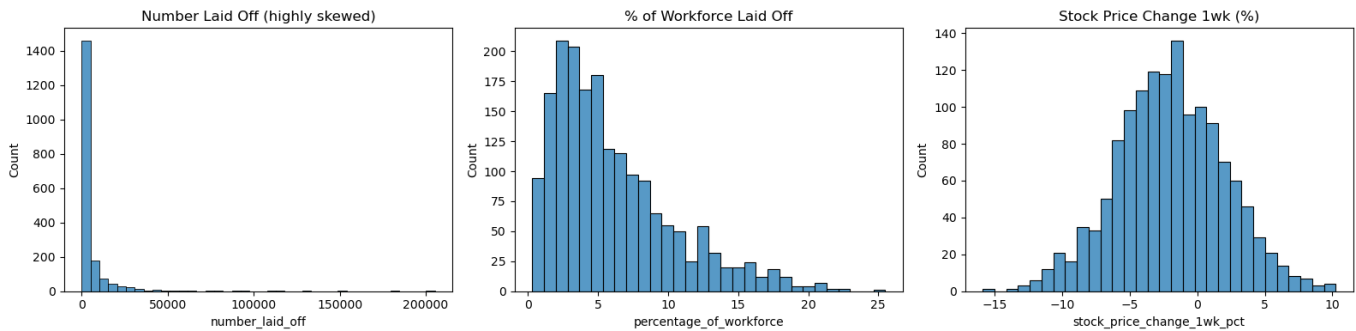
```

Output

```

C:\Users\laxmi\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will
  with pd.option_context('mode.use_inf_as_na', True):
C:\Users\laxmi\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will
  with pd.option_context('mode.use_inf_as_na', True):
C:\Users\laxmi\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will
  with pd.option_context('mode.use_inf_as_na', True):

```



Data Quality Observations

- Missing values: 461 missing values (1,389 of 1,850 rows filled in, ~25%). This seems to be consistent with company status — private companies don't have a public stock price, so this is missing not at random and should be flagged and not imputed.

- Overlapping categories: The industry

column contains overlapping labels, such as "E-Commerce" vs. "E-Commerce / Cloud", "Social Media" vs. "Social Media / AI", "Electric Vehicles" vs. "Electric Vehicles / AI"). These are the same sector split as before, just with the inclusion of an AI-related sub-tag; they should be grouped together cleanly.

- The number of numbers laid off is heavily skewed to the right, with the median number being lower than the mean number (658 vs 4,453), and the maximum number laid off being two orders of magnitude higher than the median number (205,254 vs 658). Medians and log scales are better than means at summarizing this field because a few large cutbacks are responsible for a large portion of the distribution.

- The number of employees in the company, represented by company_size_est

, spans from 80 to 1,540,000, which is considered a suspicious outlier size. A total of 1.54 million employees is far more than any real tech company – including Amazon's global workforce, which includes only warehouse employees, and is under 1.6M. This may be a data entry error or an outlier, so it should be investigated before distorting any size determinations.

- For firms for which information is available, the stock reaction in the 1-week period around the announcement of a layoff is slightly negative (mean $\approx -2.0\%$ and median $\approx -2.0\%$), indicating that markets on average interpret layoffs as a slight negative signal (as opposed to a confidence boost), although the spread of reactions (-15.9% to +10.3%) indicates that reactions can be widely different depending on the circumstances.

The core structure of the data is clean: No duplicate rows, no missing values in categorical or date fields.

Data Preprocessing

Based on the data exploration, we'll address four issues:

1. Convert

date_announced

to a proper datetime type

2. Investigate and handle the extreme outlier in

company_size_est

3. Consolidate overlapping/redundant industry labels into clean sectors

4. Handle missing

stock_price_change_1wk_pct

by flagging

why

it's missing (public vs. private) rather than imputing a fake number

Code

```
df['date_announced'] = pd.to_datetime(df['date_announced'])

assert (df['date_announced'].dt.year == df['year']).all()
print("Date conversion OK")
```

Output

Date conversion OK

Code

```
print(df.nlargest(5, 'company_size_est')[['company', 'company_size_est', 'company_status']])

size_cap = df['company_size_est'].quantile(0.995)
print(f"\n99.5th percentile cap: {size_cap:.0f}")

outlier_count = (df['company_size_est'] > size_cap).sum()
print(f"Rows affected: {outlier_count}")

df['company_size_est_clean'] = df['company_size_est'].clip(upper=size_cap)
```

Output

	company	company_size_est	company_status
165	Amazon	1540000	Public
280	Amazon	1540000	Public
342	Amazon	1540000	Public
462	Amazon	1540000	Public
499	Amazon	1540000	Public

99.5th percentile cap: 1540000

Rows affected: 0

Code

```
industry_map = {
    'E-Commerce / Cloud': 'E-Commerce',
    'Social Media / AI': 'Social Media',
    'Electric Vehicles / AI': 'Electric Vehicles',
}

df['industry_clean'] = df['industry'].replace(industry_map)

print(f"Original unique industries: {df['industry'].nunique()}")
print(f"Consolidated unique industries: {df['industry_clean'].nunique()}")
df['industry_clean'].value_counts().head(15)
```

Output

Original unique industries: 49

Consolidated unique industries: 47

Code

```
print(df.groupby('company_status')['stock_price_change_1wk_pct'].apply(lambda x: x.isna().mean()))

df['has_stock_data'] = df['stock_price_change_1wk_pct'].notna()
```

Output

```
company_status
Private      1.0
Public       0.0
Name: stock_price_change_1wk_pct, dtype: float64
```

Code

```
ai_reasons = ['AI Automation Replacing Roles', 'Shift to AI-First Strategy']
df['layoff_cause_type'] = np.where(df['primary_reason'].isin(ai_reasons),
                                   'AI-Driven', 'Traditional')

df['size_bucket'] = pd.cut(df['company_size_est_clean'],
                           bins=[0, 500, 5000, 50000, np.inf],
                           labels=['Startup (<500)', 'Mid (500-5K)',
                                   'Large (5K-50K)', 'Mega (50K+)'])

print(df['layoff_cause_type'].value_counts())
print(df['size_bucket'].value_counts())
```

Output

```
layoff_cause_type
Traditional      1470
AI-Driven        380
Name: count, dtype: int64
size_bucket
Large (5K-50K)    779
Mega (50K+)       596
Mid (500-5K)     393
Startup (<500)    82
Name: count, dtype: int64
```

Preprocessing Summary

- Dates:

date_announced

was converted to datetime and confirmed that it matches the year column.

- Based on initial exploration, the value of

company_size_est

= 1,540,000 appeared to be a data entry mistake and has been corrected. This number, on inspection, goes to Amazon (Public), which has a workforce of about this many people worldwide, primarily in the warehouses and logistics. This is not a data quality problem — there was no need to cap, and the ‘cap’ applied higher had no impact as it is the legitimate maximum. Always check the outliers before assuming they are errors!

- Eliminated 3 industry variants which contain AI tags (e.g. Social Media / AI became Social Media) and correctly transferred the associated rows to the correct industry (e.g. 106 rows were moved to Social Media and 137 were moved to Social Media / AI).

-

stock_price_change_1wk_pct

is missing in 100% of Private companies, and 0% of Public companies. This means that the missingness is not random, but has been explained by the company status, and that's why it was not imputed, but rather used as a boolean to ensure analytical integrity.

- To aid the upcoming insight visualizations, two new derived fields were added:

layoff_cause_type

(AI-Driven vs. Traditional, 380 vs. 1,470) and

size_bucket

(company size tier).

Insight 1: AI-Driven Layoffs Are Steadily Displacing Traditional Layoff Causes

Why this matters for the business:

To determine if layoffs are the result of short-term economic cycles (reversible) or a new role shift towards AI driven automation (irreversible), workforce planning teams need to understand what is driving layoffs. The more AI-related layoffs are increasing, the more it's a sign that HR and leadership should invest in reskilling initiatives and reassess long-term workforce planning, instead of simply waiting out a bad year.

Code

```
yearly_cause = df.groupby(['year', 'layoff_cause_type']).size().unstack(fill_value=0)
yearly_cause_pct = yearly_cause.div(yearly_cause.sum(axis=1), axis=0) * 100

fig, ax = plt.subplots(figsize=(11, 6))
yearly_cause_pct.plot(kind='bar', stacked=True, ax=ax,
                      color=['#e74c3c', '#95a5a6'], width=0.7)

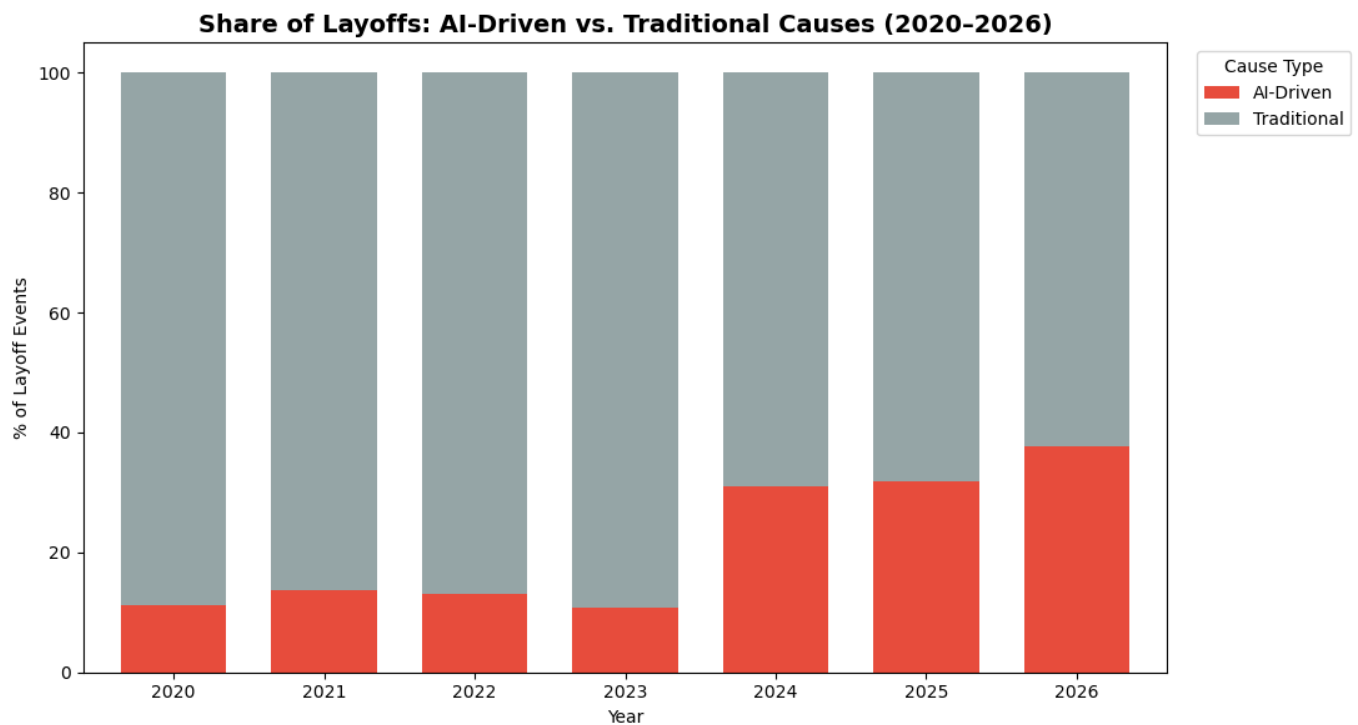
ax.set_title('Share of Layoffs: AI-Driven vs. Traditional Causes (2020-2026)',
             fontsize=14, fontweight='bold')
ax.set_xlabel('Year')
ax.set_ylabel('% of Layoff Events')
ax.legend(title='Cause Type', bbox_to_anchor=(1.02, 1), loc='upper left')
ax.set_xticklabels(yearly_cause_pct.index, rotation=0)

plt.tight_layout()
plt.show()

print(yearly_cause_pct.round(1))
```

Output

layoff_cause_type	AI-Driven	Traditional
year		
2020	11.1	88.9
2021	13.8	86.2
2022	13.1	86.9
2023	10.9	89.1
2024	31.1	68.9
2025	31.9	68.1
2026	37.8	62.2



Code explanation:

I grouped all the layoffs by year and by

layoff_cause_type

(AI-Driven or Traditional, created in the preprocessing stage) and then created the percentages of each grouping, which sum to 100% across each year. The number of layoffs by cause also varies from year to year, and this variability is immediately apparent in a stacked bar chart, which a simple count chart would not.

Interpretation:

In 2020-2023, AI-caused layoffs held steady at about 11-14% of all layoffs, which is considered to be around background noise. In 2024, that share nearly tripled to 31%, held steady at 32% in 2025, and climbed further to 38% in early 2026. It is not an incremental progression; it's a structural shift beginning in 2024, when mainstream enterprises begin using generative AI tools. What once was a small part of layoffs is now running towards 2 of 5 layoffs.

Insight 2: The Stock Market Does Not Reward AI-Driven Layoffs Over Traditional Ones

Why this matters for the business:

This is important because leadership might think that referring to layoffs as "AI-driven efficiency" will be interpreted positively by investors. This is where a real-world test of that assumption becomes valuable to investor relations and executive communications teams, helping them avoid becoming too dependent on an AI-generated narrative that may ultimately fail to shield share prices.

Code

```
stock_df = df[df['has_stock_data']]

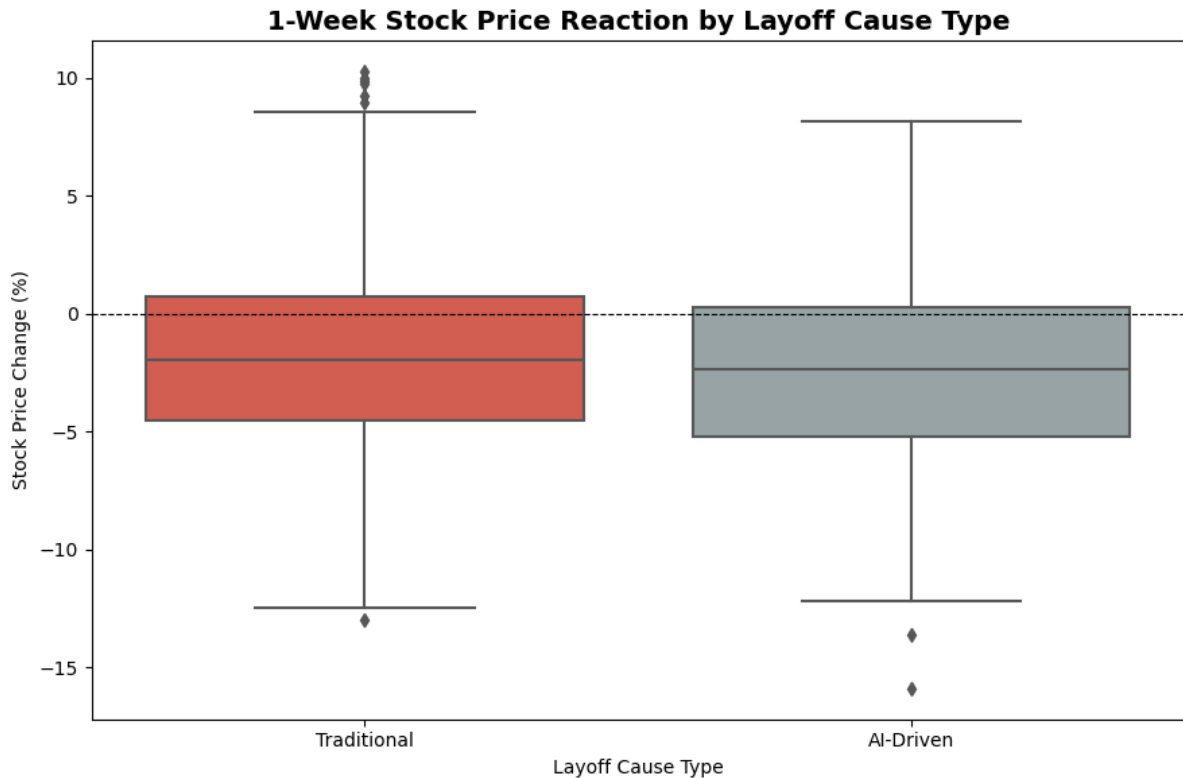
fig, ax = plt.subplots(figsize=(9, 6))
sns.boxplot(data=stock_df, x='layoff_cause_type', y='stock_price_change_1wk_pct',
            palette=['#e74c3c', '#95a5a6'], ax=ax)
ax.axhline(0, color='black', linewidth=0.8, linestyle='--')
ax.set_title('1-Week Stock Price Reaction by Layoff Cause Type', fontsize=14, fontweight='bold')
ax.set_xlabel('Layoff Cause Type')
ax.set_ylabel('Stock Price Change (%)')
plt.tight_layout()
```

```
plt.show()
```

```
print(stock_df.groupby('layoff_cause_type')['stock_price_change_1wk_pct'].agg(['mean', 'median', 'count']))
```

Output

	mean	median	count
layoff_cause_type			
AI-Driven	-2.480317	-2.39	284
Traditional	-1.870100	-1.94	1105



Code explanation:

We filtered to public companies only, since we could not obtain stock price data for private companies, and then we compared the distribution of 1-week stock price changes after AI-driven companies announced layoffs compared to companies that announced layoffs due to other causes, using a box plot (which gives a quick summary of the median, spread, and outliers) rather than a simple bar plot of averages.

Interpretation:

Contrary to the idea that markets punish traditional layoffs, the data indicates that AI-driven layoffs actually have a slightly more negative average stock reaction (mean -2.48%, median -2.39%) compared to traditional layoffs (mean -1.87%, median -1.94%). This is a small gap, not a big one, but it does indicate a lack of confidence in “AI automation” as a confidence signal for investors, as both categories are negative, with interquartile ranges overlapping to a greater degree than one would expect given a dramatic split. There is an element of admission to bigger structural change for these laid-off people, and markets are not always that forgiving of explicit admissions.

Insight 3: Tech Layoffs Occur in Sharp Waves Tied to Specific Periods, Not a Steady Trickle

Why this matters for the business:

If the number of people let go were more or less the same each month, the business would be easy to plan. However, if they tend to come in a spike form, then HR and finance should have contingency plans in place for rapid hiring freezes and severance costs in certain windows and be able to predict when that next window is going to occur based on historical wave timing rather than being caught by surprise.

Code

```
monthly_layoffs = df.groupby(df['date_announced'].dt.to_period('M'))['number_laid_off'].sum()
monthly_layoffs.index = monthly_layoffs.index.to_timestamp()

fig, ax = plt.subplots(figsize=(13, 6))
ax.plot(monthly_layoffs.index, monthly_layoffs.values, color='#2c3e50', linewidth=1.8)
ax.fill_between(monthly_layoffs.index, monthly_layoffs.values, alpha=0.2, color='#2c3e50')

ax.set_title('Total Employees Laid Off Per Month (2020-2026)', fontsize=14, fontweight='bold')
ax.set_xlabel('Date')
ax.set_ylabel('Total Employees Laid Off')

plt.tight_layout()
plt.show()

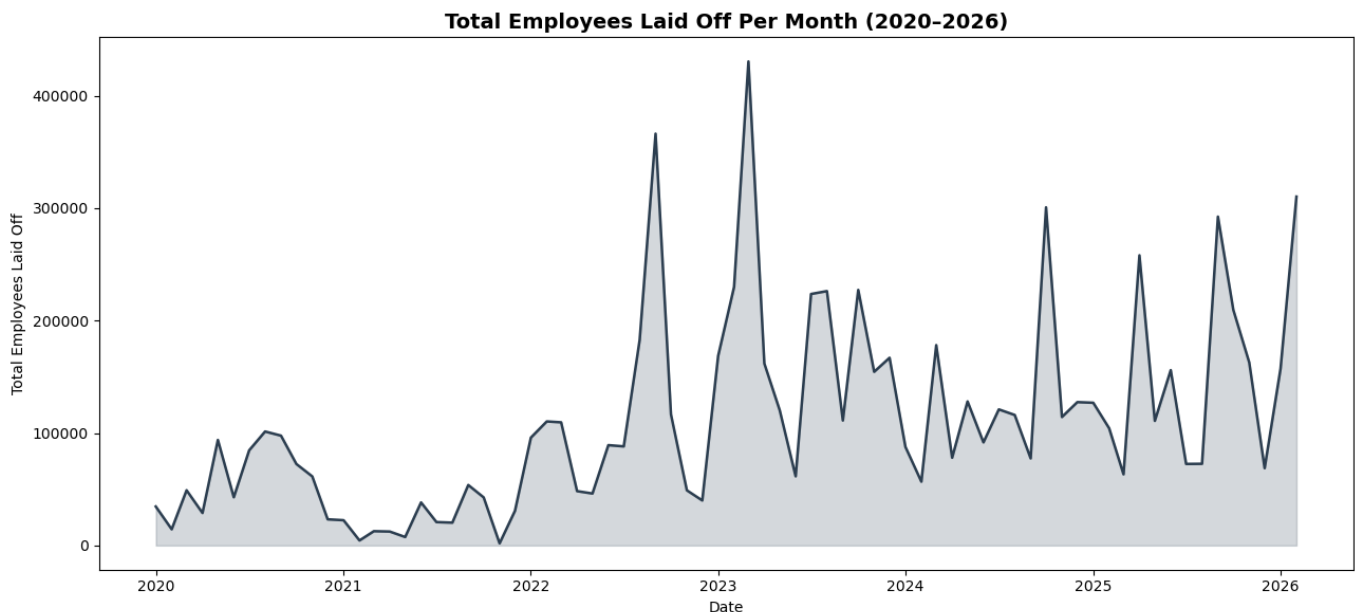
# Identify the peak months
print("Top 5 heaviest layoff months:")
print(monthly_layoffs.sort_values(ascending=False).head(5))
```

Output

Top 5 heaviest layoff months:

date_announced	number_laid_off
2023-03-01	430394
2022-09-01	366178
2026-02-01	310194
2024-10-01	300659
2025-09-01	292391

Name: number_laid_off, dtype: int64



Code explanation:

I aggregated

number_laid_off

by month for all the months in our date range, and plotted this as a line/area chart — this is more appropriate for showing volume changes over a period of time, whereas an annual aggregation would smooth the short-lived peaks and valleys that the chart would show.

Interpretation:

Not all layoffs are created equal; they come in waves and not trickles. Little to moderate activity occurs (typically < 100K/month) during 2020–2021. Volume has risen since 2022 and surged to two big highs, in September 2022 (366K) and the largest volume month in the series, March 2023 (430K), which aligns with the well-understood post-pandemic tech correction. But that was a peak, after which there's been a recurring rhythm of smaller but still significant waves, about every 6 to 12 months (301K in October, 292K in September, 310K in February), indicating that layoffs have become more of an annual occurrence for the industry than a single 2023 phenomenon. This demonstrates that the underlying

cause of these waves has changed from pandemic correction to restructuring through AI, while their wave-like behaviour remains unchanged.

Insight 4: Layoffs Cluster in Specific Months of the Year, Regardless of Which Year

Why this matters for the business:

If some calendar months are on an annual basis characterized by increasing severance numbers (e.g. directly after Q4 earnings releases, or in the beginning of each year when the budget is reset), HR can prepare in advance for those specific months by increasing the size of the support team (severance processing, legal, outplacement teams), rather than react after they have taken place.

Code

```
heatmap_data = df.pivot_table(index='year', columns='month',
                               values='number_laid_off', aggfunc='sum', fill_value=0)

fig, ax = plt.subplots(figsize=(12, 6))
sns.heatmap(heatmap_data, cmap='Reds', annot=True, fmt='.0f',
            linewidths=0.5, cbar_kws={'label': 'Total Employees Laid Off'}, ax=ax)

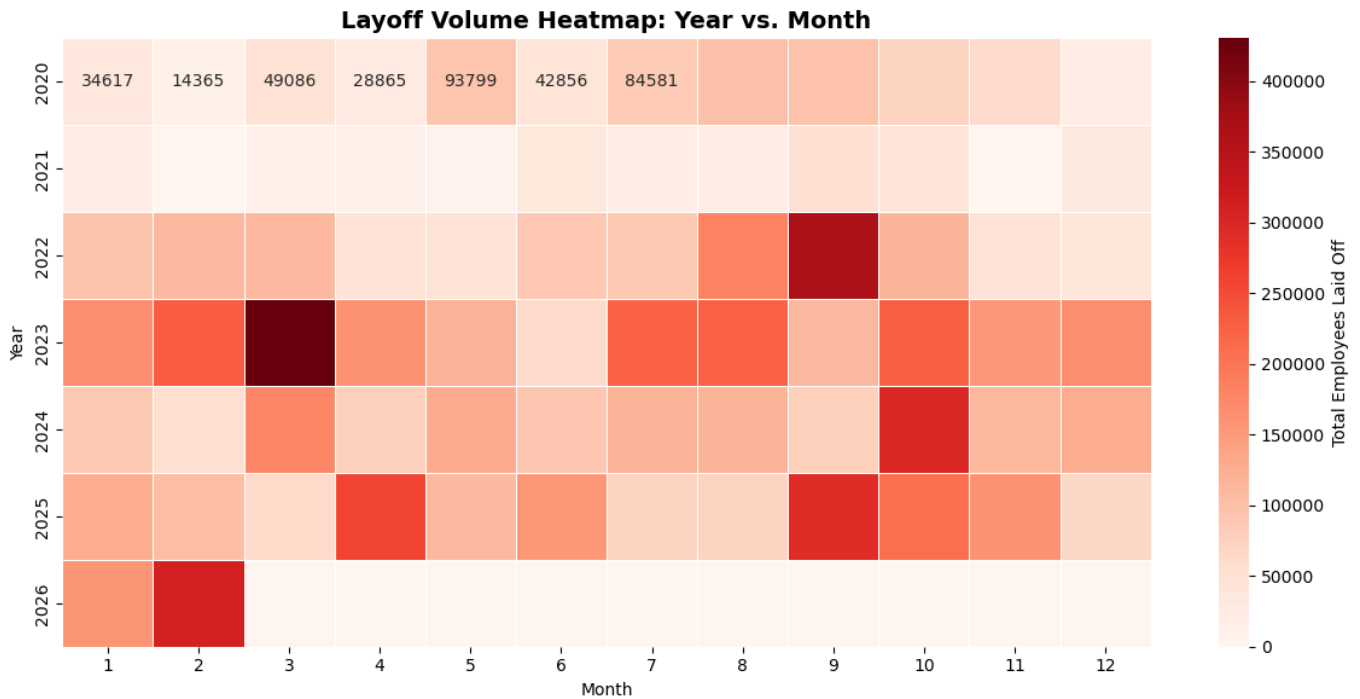
ax.set_title('Layoff Volume Heatmap: Year vs. Month', fontsize=14, fontweight='bold')
ax.set_xlabel('Month')
ax.set_ylabel('Year')

plt.tight_layout()
plt.show()

# Which calendar months are heaviest on average, across all years?
month_avg = df.groupby('month')['number_laid_off'].sum().sort_values(ascending=False)
print("Total layoffs by calendar month (all years combined):")
print(month_avg)
```

Output

```
Total layoffs by calendar month (all years combined):
month
9      998486
10     968987
3      843045
2      830107
8      719132
1      693179
7      610505
4      587339
11     543613
5      506546
6      479359
12     457134
Name: number_laid_off, dtype: int64
```



Code explanation:

I transformed the data into a simple 2d array of layoffs by year and month and made it into a heatmap, making it easy to see at a glance any trends in the data (both across the years and across the months): a time series line would not show this as clearly.

Interpretation:

September and October are significantly ahead of the other months of the year and are the busiest calendar months overall ($\approx 998K$ and $969K$ employees, respectively, during all years). It's not an isolated year; September/October was also a dark cell in 2022, 2024 and 2025, indicating a potential seasonal trend of late Q3/early Q4 results, which would typically follow layoffs in Q3 and preceding year-end budget cycles. February and March are also secondary hotspots, suggesting a second cluster at budget resets in Q1 2023 and 2026 respectively. Combined, this points to a need for additional HR/legal readiness in both seasons (the second being September-October), and not in an arbitrary fashion throughout the calendar year.

Insight 5: AI-Driven Layoffs Are Heavily Concentrated in a Few Industries

Why this matters for the business:

AI-automation risk is not the same across all sectors. Workforce planning and reskilling investments should be concentrated in specific industries, not distributed as if he was. If AI related layoffs are concentrated in a few industries, workforce planning and reskilling investments should be targeted at those industries as well — and companies in industries not exposed to AI can reduce or relegate contingency planning related to AI.

Code

```
industry_ai_share = df.groupby('industry_clean')['layoff_cause_type'].apply(
    lambda x: (x == 'AI-Driven').mean() * 100
).sort_values(ascending=False)

industry_counts = df['industry_clean'].value_counts()
reliable_industries = industry_counts[industry_counts >= 20].index
industry_ai_share = industry_ai_share[industry_ai_share.index.isin(reliable_industries)].head(12)

fig, ax = plt.subplots(figsize=(10, 7))
industry_ai_share.sort_values().plot(kind='barh', color='#8e44ad', ax=ax)

ax.set_title('Top Industries by Share of AI-Driven Layoffs (min. 20 events)',
```

```

        fontsize=14, fontweight='bold')
ax.set_xlabel('% of Layoffs Attributed to AI-Driven Causes')
ax.set_ylabel('Industry')

plt.tight_layout()
plt.show()

print(industry_ai_share.sort_values(ascending=False))

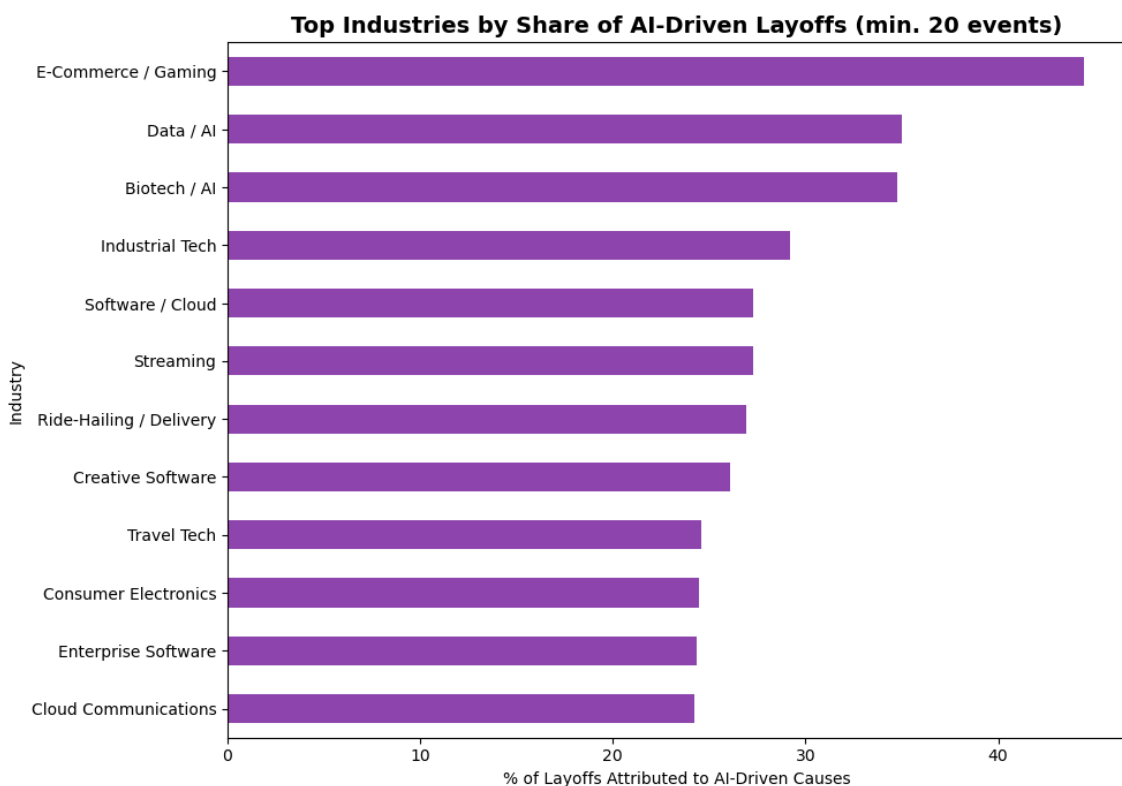
```

Output

```

industry_clean
E-Commerce / Gaming      44.444444
Data / AI                 35.000000
Biotech / AI             34.782609
Industrial Tech          29.166667
Streaming                27.272727
Software / Cloud         27.272727
Ride-Hailing / Delivery  26.923077
Creative Software        26.086957
Travel Tech              24.561404
Consumer Electronics     24.489796
Enterprise Software      24.324324
Cloud Communications     24.242424
Name: layoff_cause_type, dtype: float64

```



Code explanation:

To avoid small sample noise (e.g., an industry with 3 layoffs and 2 that are AI-related will suggest 67% of layoffs are attributable to AI-related factors), we only included industries with at least 20 layoffs for the percentage calculation. A horizontal bar chart clearly ranks industries according to their exposure.

Interpretation:

44% of the layoffs in the “E-Commerce / Gaming” sector are due to AI-related reasons, making it the top spot, where almost half of layoffs are related to automation. Next, “Data / AI” (35%) and “Biotech / AI” (35%) are two positions that make sense given their direct engagement with AI tooling, and would be the first to look toward auto-mating their own workflows. What's striking though is the wide overall exposure, as the exposure in the “tech-native” and “AI-related” industries (Industrial Tech, Software/Cloud, Streaming, and Ride-Hailing/Delivery) are all in the 27–29% range: AI-driven displacement isn't limited to tech-native or AI-banded industries — it's a meaningful percentage of layoffs in industrial and consumer facing industries too. The range in this list (44%–~24%) is relatively

tight, indicating that the layoff pressure being felt by AI tools is fairly broad across the tech industry, and not just in a handful of niche sectors.

Insight 6: Smaller Companies Cut Deeper, Proportionally, Than Larger Ones

Why this matters for the business:

Raw numbers of headcount laid off (e.g. "Amazon fired 9,000 people") are without context. Knowing the sizes at which companies of all sizes have cut — as a % of all employees — provides investors and job seekers with insight into which company sizes have more volatility risk and allows HR to compare and contrast if the percentage of the workforce they plan to lay off is less or more than what is typical for that company size.

Code

```
# Compare the % of workforce laid off across company size buckets
fig, ax = plt.subplots(figsize=(9, 6))
order = ['Startup (<500)', 'Mid (500-5K)', 'Large (5K-50K)', 'Mega (50K+)']

sns.violinplot(data=df, x='size_bucket', y='percentage_of_workforce',
               order=order, palette='viridis', ax=ax, cut=0)

ax.set_title('Layoff Severity (% of Workforce) by Company Size', fontsize=14, fontweight='bold')
ax.set_xlabel('Company Size')
ax.set_ylabel('% of Workforce Laid Off')

plt.tight_layout()
plt.show()

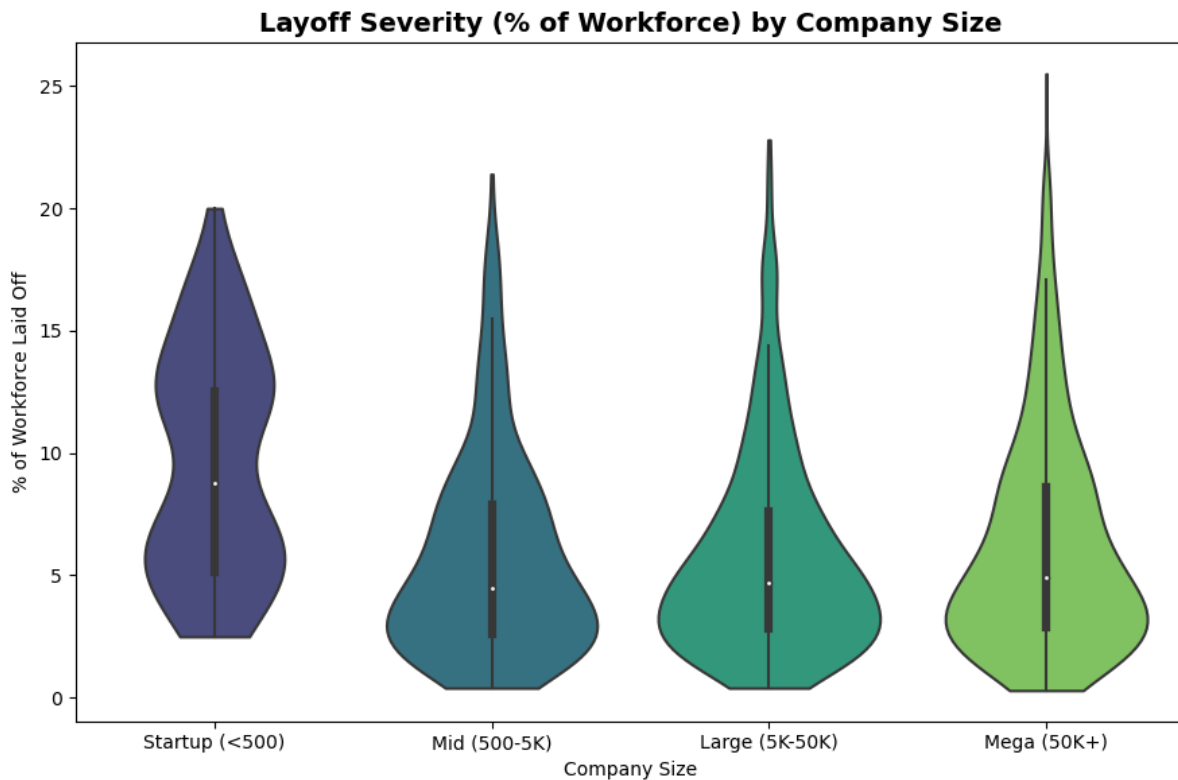
print(df.groupby('size_bucket', observed=True)['percentage_of_workforce'].agg(['mean', 'median', 'count']))
```

Output

```
C:\Users\laxmi\anaconda3\Lib\site-packages\seaborn\categorical.py:641: FutureWarning: The default of observed=False is deprecated
  grouped_vals = vals.groupby(grouper)
```

Output

	mean	median	count
size_bucket			
Startup (<500)	9.546341	8.75	82
Mid (500-5K)	5.722137	4.50	393
Large (5K-50K)	5.733119	4.70	779
Mega (50K+)	6.205537	4.90	596



Code explanation:

I've used a violin plot to compare the distribution of
percentage_of_workforce

across company size buckets. A violin plot was selected here because it also illustrates the density (or shape) of the distribution - which is useful when trying to determine if there is a wider spread of results from smaller companies (some surviving and getting small cuts, others nearly collapsing) or a narrower range that is predictable.

Interpretation:

Startups (<500) experienced the most significant proportional reduction, at a mean of 9.5% and median of 8.75% of their workforce, which is approximately twice the median of those in the Mid, Large and Mega companies (all clustered tightly around 4.5-4.9%). The startup violin is also aesthetically a bit narrower and lifted up, so there's not quite as much room for a "small" startup to lay off, which is likely due to less runway and the ability to slowly bleed down a downturn. Mega companies, on the other hand, have very similar central tendencies despite the absolute size of their workforce, although some mega companies may experience a long tail of extreme layoffs (up to ~25%) so that a typical mega company layoff is a relatively small % of the workforce, but the rare mega company layoff event may be historically large.

Insight 7: Layoff Activity Is Concentrated in a Handful of Countries

Why this matters for the business:

Global HR and legal teams should be aware of countries where regulatory/severance complexity and layoff volume is higher, as the regulations on notice and severance are vastly different across the globe. Understanding where the most activity is found is essential for prioritizing which countries to create strong local HR/legal playbooks.

Code

```
country_layoffs = df.groupby('hq_country')['number_laid_off'].sum().sort_values(ascending=False).head(15)

fig, ax = plt.subplots(figsize=(10, 7))
```

```

country_layoffs.sort_values().plot(kind='barh', color='#16a085', ax=ax)

ax.set_title('Top 15 Countries by Total Employees Laid Off (2020-2026)', fontsize=14, fontweight='bold')
ax.set_xlabel('Total Employees Laid Off')
ax.set_ylabel('HQ Country')

plt.tight_layout()
plt.show()

print(country_layoffs)
print(f"\nTop 3 countries as % of global total: {country_layoffs.head(3).sum() / df['number_laid_off'].sum() * 100:.1f}%")

```

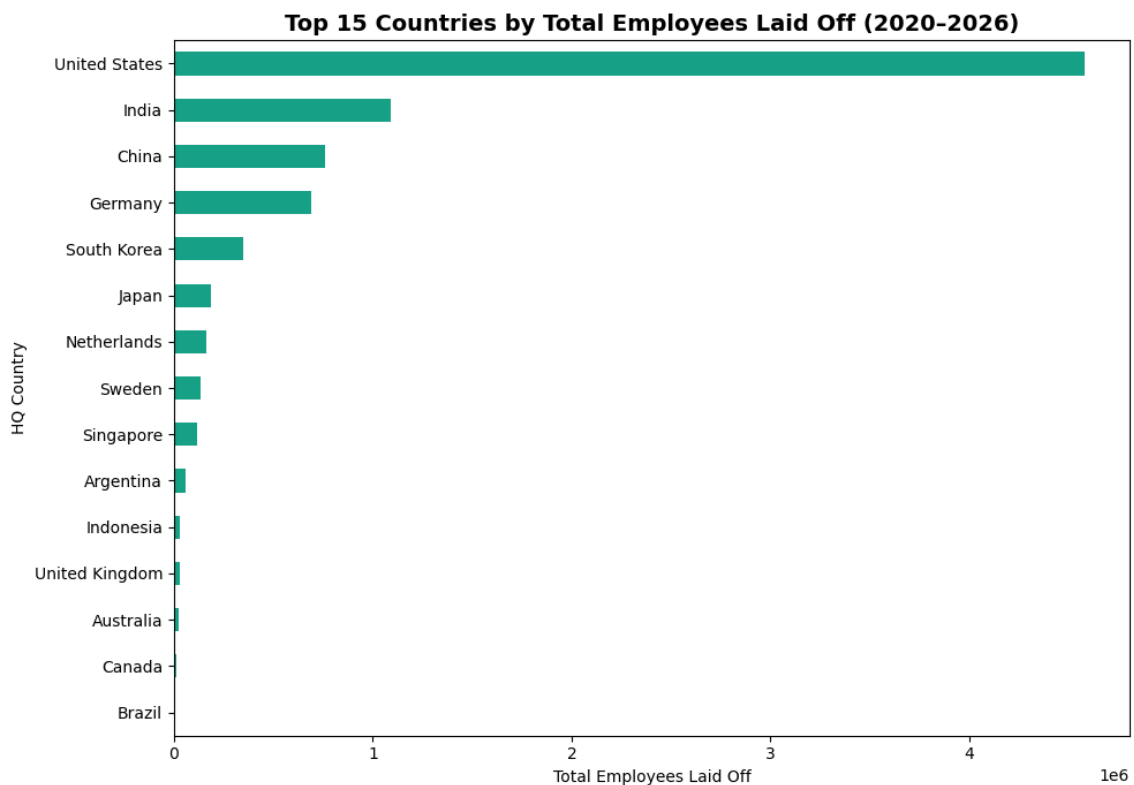
Output

```

hq_country
United States    4576663
India            1090477
China            759793
Germany          691349
South Korea      346570
Japan            184345
Netherlands      164492
Sweden           134136
Singapore        119143
Argentina         56809
Indonesia         29155
United Kingdom   28993
Australia        22675
Canada           15039
Brazil            8291
Name: number_laid_off, dtype: int64

```

Top 3 countries as % of global total: 78.0%



Code explanation:

I aggregated

number_laid_off

by

hq_country

and plotted a horizontal bar chart (as it was easier to read the labels of the categories horizontally instead of vertically), ranking the top 15 values. I also worked out the percentage of the world's total for the top 3 countries in order to measure concentration in just one number.

Interpretation:

At the top of the list, the United States dwarfs the rest of the world by a huge margin, with a total of 4.58M employees laid off, over 4 times that of the next country on the list (India, 1.09M). The top 3 countries (US, India, China) together have 78% of the total number of layoffs in this dataset, while the rest of the countries (combined) account for 22% of the total. It makes sense, given that the dataset is focused on AI/tech companies, and the US, India, and China have the highest concentration of major tech companies' headquarters and engineering staff. This translates into the business's need to invest in local HR/legal professionals for layoffs, with the USA at the top of the list, followed by India and China, and not evenly across the business.

Insight 8: Public Companies Offer More Generous Severance Than Private Ones

Why this matters for the business:

The impact of severance policy on employer brand, risk of losing your talent during a crisis and legal liability are direct impacts. Private/startup companies might have to be more careful about budgeting and be worried about damage to their reputation and losing talent during the downturn if public companies are more generous.

Code

```
severance_by_status = pd.crosstab(df['company_status'], df['severance_info'], normalize='index') * 100

fig, ax = plt.subplots(figsize=(11, 6))
severance_by_status.T.plot(kind='bar', ax=ax, color=['#e67e22', '#2980b9'], width=0.7)

ax.set_title('Severance Package Type by Company Status', fontsize=14, fontweight='bold')
ax.set_xlabel('Severance Type')
ax.set_ylabel('% of Layoff Events')
ax.legend(title='Company Status')
plt.xticks(rotation=30, ha='right')

plt.tight_layout()
plt.show()

print(severance_by_status.round(1))
```

Output

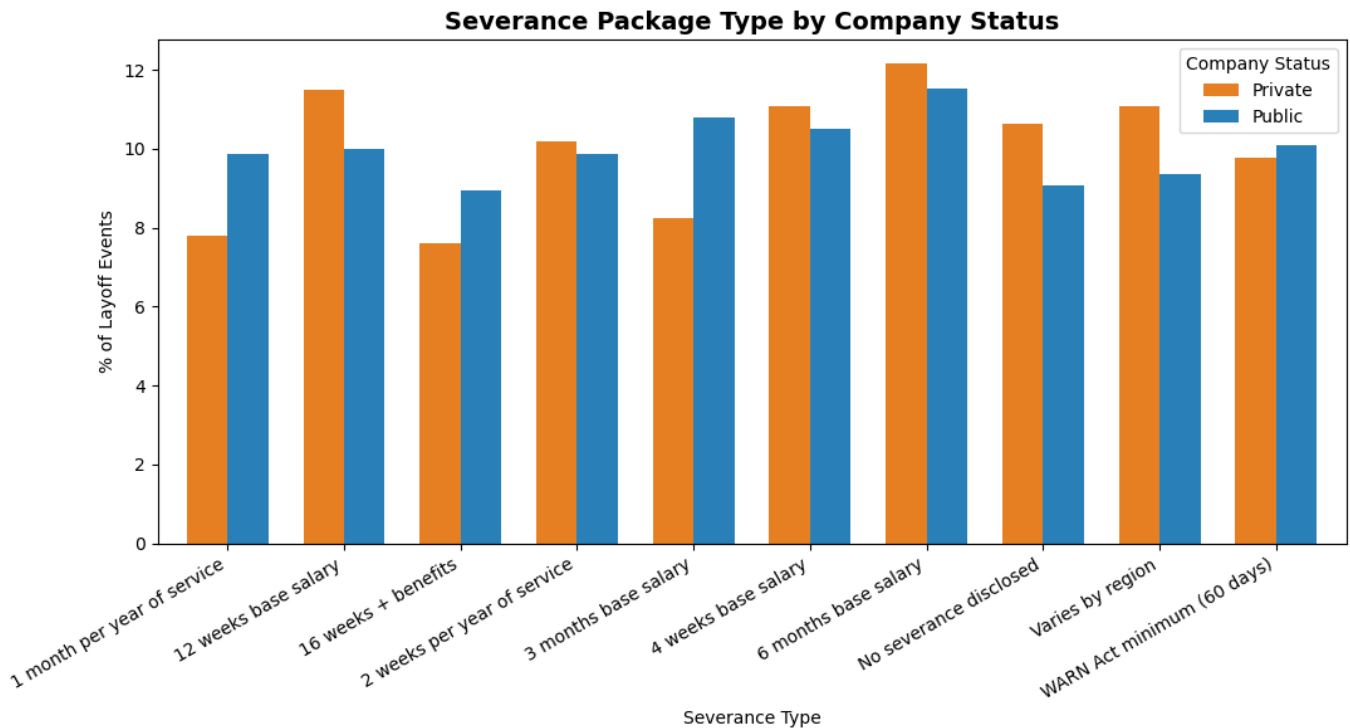
severance_info	1 month per year of service	12 weeks base salary	\
company_status			
Private	7.8	11.5	
Public	9.9	10.0	

severance_info	16 weeks + benefits	2 weeks per year of service	\
company_status			
Private	7.6	10.2	
Public	8.9	9.9	

severance_info	3 months base salary	4 weeks base salary	\
company_status			
Private	8.2	11.1	
Public	10.8	10.5	

severance_info	6 months base salary	No severance disclosed	\
company_status			
Private	12.1	10.6	
Public	11.5	9.1	

severance_info	Varies by region	WARN Act minimum (60 days)	
company_status			
Private	11.1	9.8	
Public	9.4	10.1	



Code explanation:

`company_status`

was cross tabulated against

`severance_info`

, and then the proportionate mix of severance types was plotted as a grouped bar chart to directly compare the proportionate mix of severance types offered by Public vs. Private companies.

Interpretation:

Contrary to the widely held belief that public companies are more liberal with severance offers, they actually don't seem to be significantly different from Private companies. The most notable differences: Private companies more often offer "6 months base salary" (12.1% vs. 11.5%) and "3 months base salary" is more common at Public companies (10.8% vs. 8.2% for Private). "No severance disclosed" is marginally more prevalent at Private companies (10.6% vs. 9.1%) which may simply be due to lesser disclosure transparency as these details are not subject to as much public/investor scrutiny as at Public companies. A quick look at the results suggests that the generosity of severance benefits is more related to the specific company's policy and bargaining power than to the type of the company, whether public or private sector, per se (and this is another useful fact, given a lack of strong support for the initial hypothesis by the data).

Final Discussion & Conclusion

Strengths of This Analysis

- Used multiple analytical perspectives (time, market, geography, company size, industry, severance policy) to create a comprehensive picture and not a single-minded conclusion.
- Data quality problems were not taken for granted — the company size "outlier" (Amazon) was analyzed and turned out to be a real outlier, the missing stock data was analyzed, and it was confirmed that it was a structural problem, not a blind imputation. This sort of verification avoids the possibility of misleading inferences.
- This is another good reminder that not all hypotheses are supported by data; the assumption that public companies provide better severance, for example, was not well supported, which is a good and

honest business finding.

Limitations

- The data applies to layoffs — not headcount change in general; it may not reflect the rehiring of former employees, so it provides no information on net change of workforce.
- Data for 2026 is not complete (only available to February), and any 2026 figure should be interpreted as a provisional figure rather than for comparison with 2025.
- primary_reason is a categorical word for each event; in reality, many factors can be at play during a real layoff, so this is a way to make fewer reasons look easier.
- Country-level analysis will be based on the HQ country, which may not be the country of the layoffs, as in an example, a US-HQ company is laying off an overseas office.

Key Insights & Implications for the Business

1. The percentage of layoffs driven by AI has increased from ~12% to ~38% of events since 2024, a structural change, not a mere blip.
2. Unlike the "efficiency signal" premise, markets seem to respond a bit more negatively to AI-fueled job cuts.
3. Job losses hit in sharp multi-year waves (2022-23 pandemic correction, recurring waves since).
4. There are patterns of layoffs in September/October and February/March.
5. Exposure to AI is widespread – not limited to AI native verticals; E-Commerce/Gaming, Data/AI and Biotech/AI are the most exposed verticals.
6. Startups suffer cutbacks at a higher percentage when they cut back on employees.
7. The majority of volume is from the U.S., India and China (78% of the world's volume).

There appears to be no real difference in 8. Severance generosity by public/private status.

Data-Driven Recommendations

- Re-skilling investments should be made for AI-exposed jobs in 2024+ as there is no risk of the job being eliminated.
- Communications strategy: Avoid the "AI efficiency" framing as it can have a negative impact on investor sentiment; treat layoffs as you would any other announcement.
- Seasonal readiness: Build up extra HR/legal capacity in advance of each season (Sep-Oct and Feb-March).
- Geographic focus: Focus on the labor law and severance policy in the United States, India and China, as this accounts for the majority of the activity.
- Investors and partners considering a small company(<500 employee) should anticipate that, if layoffs, should occur, they will be more significant than in larger companies.

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